Large-Scale and Multi-Structured Databases

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DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE

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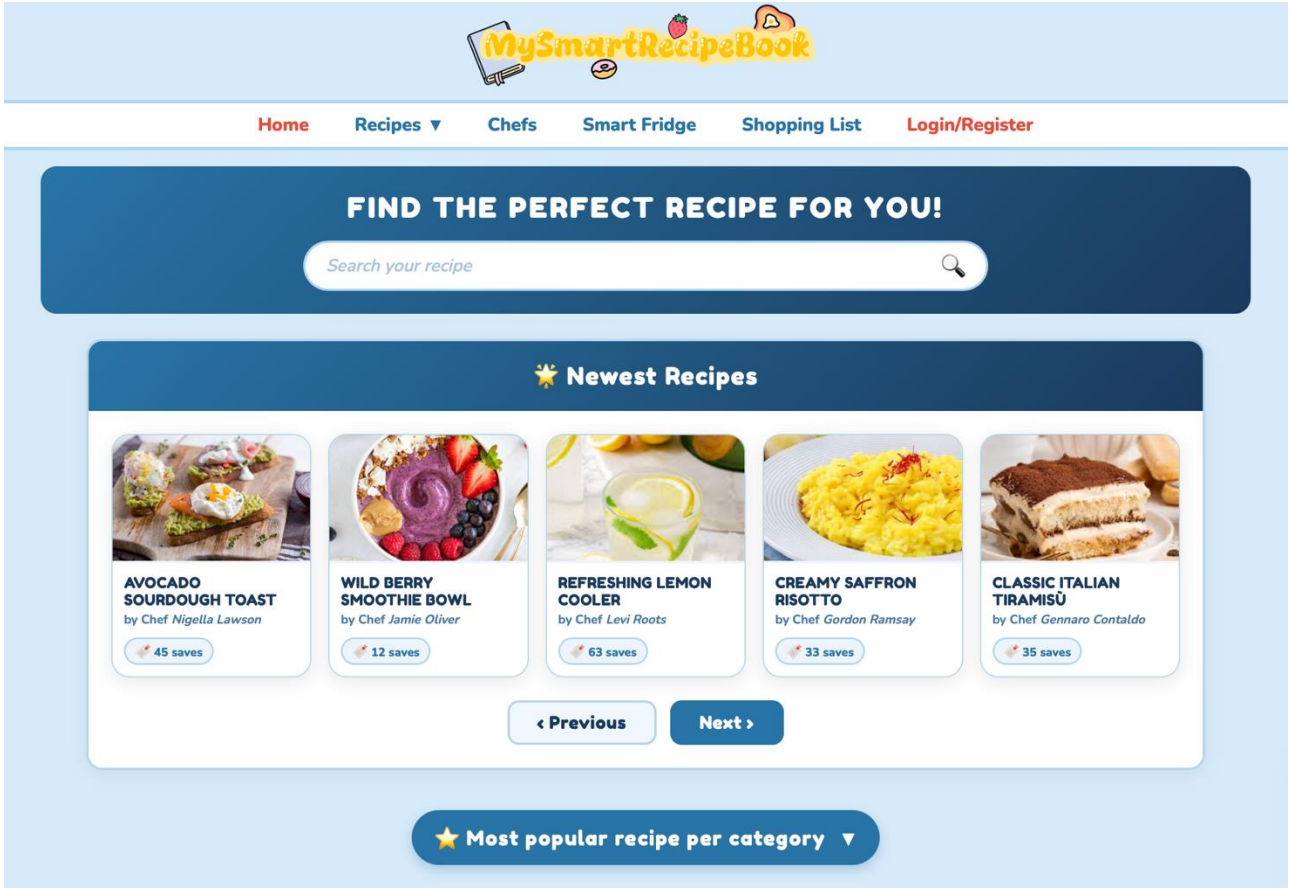
Application Highlights

MySmartRecipeBook is a recipe based web application designed to discover professional recipes and help you reducing food waste!

- **Advanced Search**: Users can browse through thousands of recipes using detailed filters such as category, or specific chefs, and save their favorite.
- **Smart Fridge**: This feature allows users to input available ingredients from their pantry to receive recipe recommendations, promoting culinary creativity and reducing food waste.
- **Smart Shopping List**: The application includes a portable digital shopping list that allows users to keep track of necessary ingredients while at the supermarket.
- **Application Analytics**: The platform provides insights into culinary trends, enabling the Admin to track monthly registrations, monitor the most popular chefs, and identify trending recipe categories.
- **Certified Content**: A dedicated administrative layer ensures a safe environment by verifying chefs' registration request and reviewing all submitted recipes before publication to maintain high-quality standards.

Actors and Mockups (i)

Unregistered User



The homepage is designed with a search bar for finding recipes via title or substring matching, and it immediately presents users with the newest recipes available in the application.

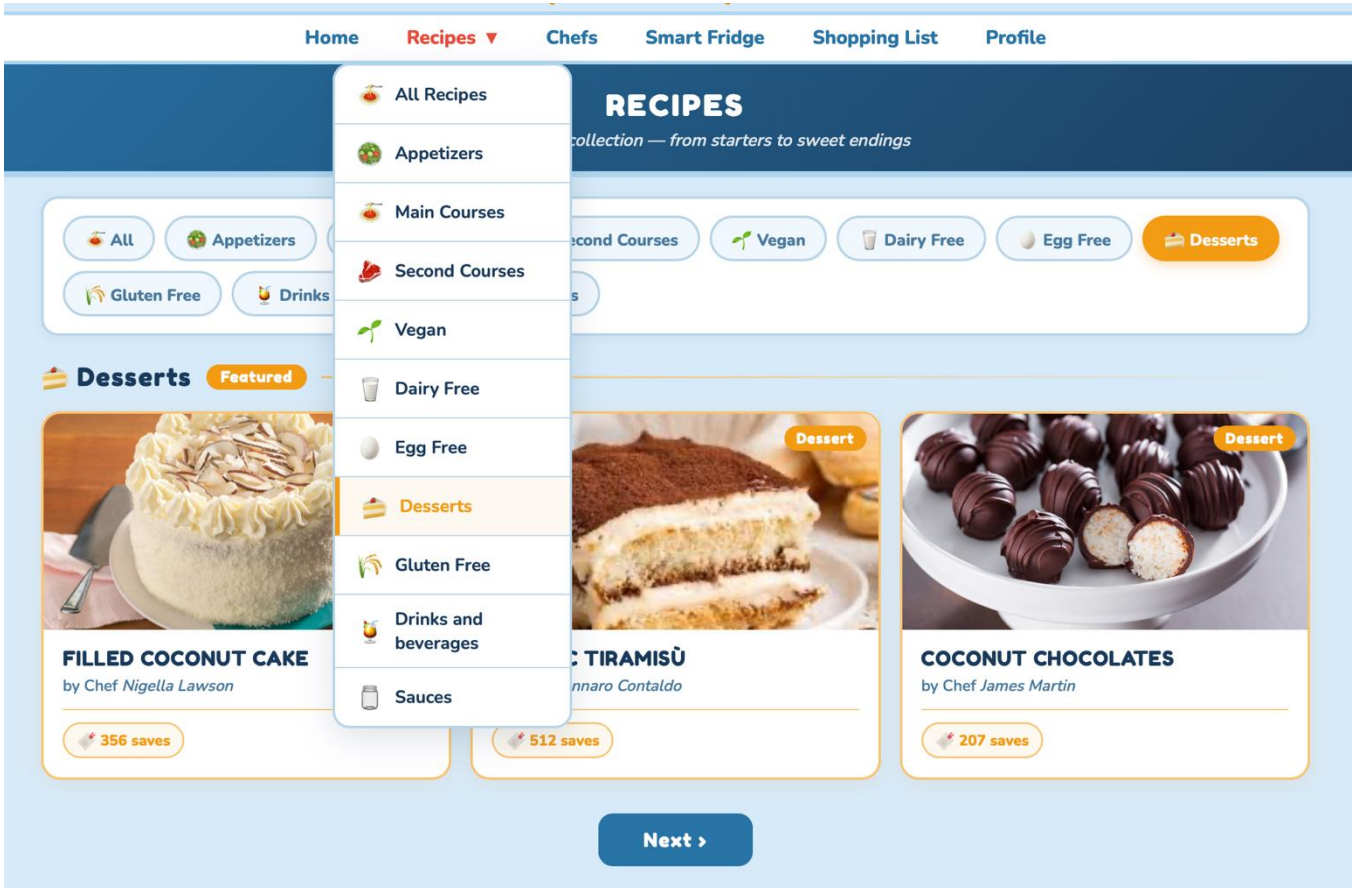
Actors and Mockups (ii)

Unregistered User

By clicking the dedicated button on the homepage, users can instantly view the most popular recipe for each available category.

Actors and Mockups (iii)

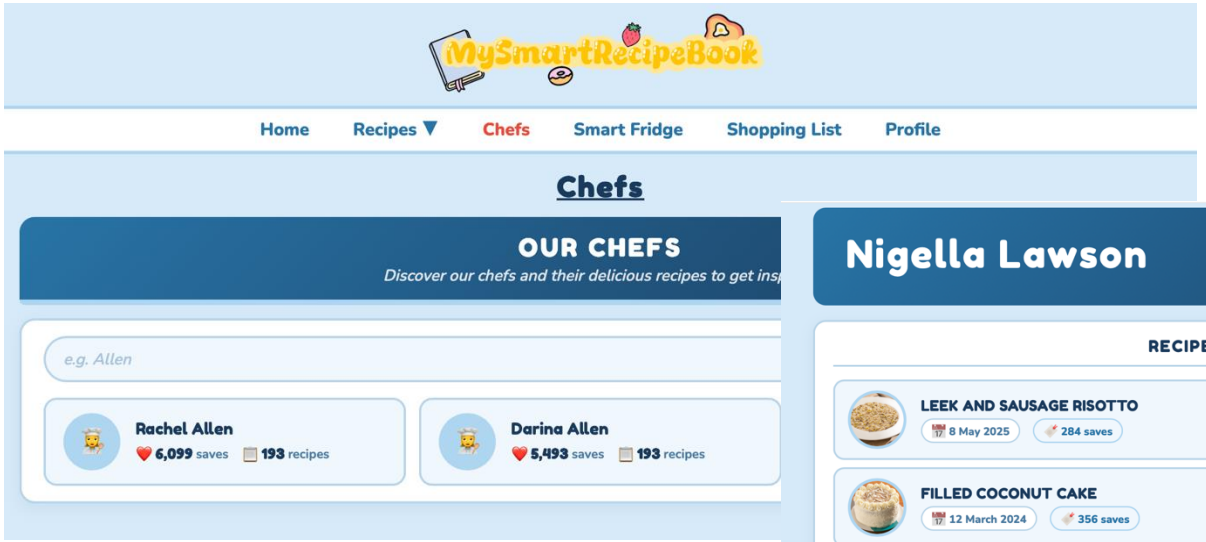
Unregistered User



Any user can search recipes, browse categories and view complete recipe details before registering.

Actors and Mockups (iv)

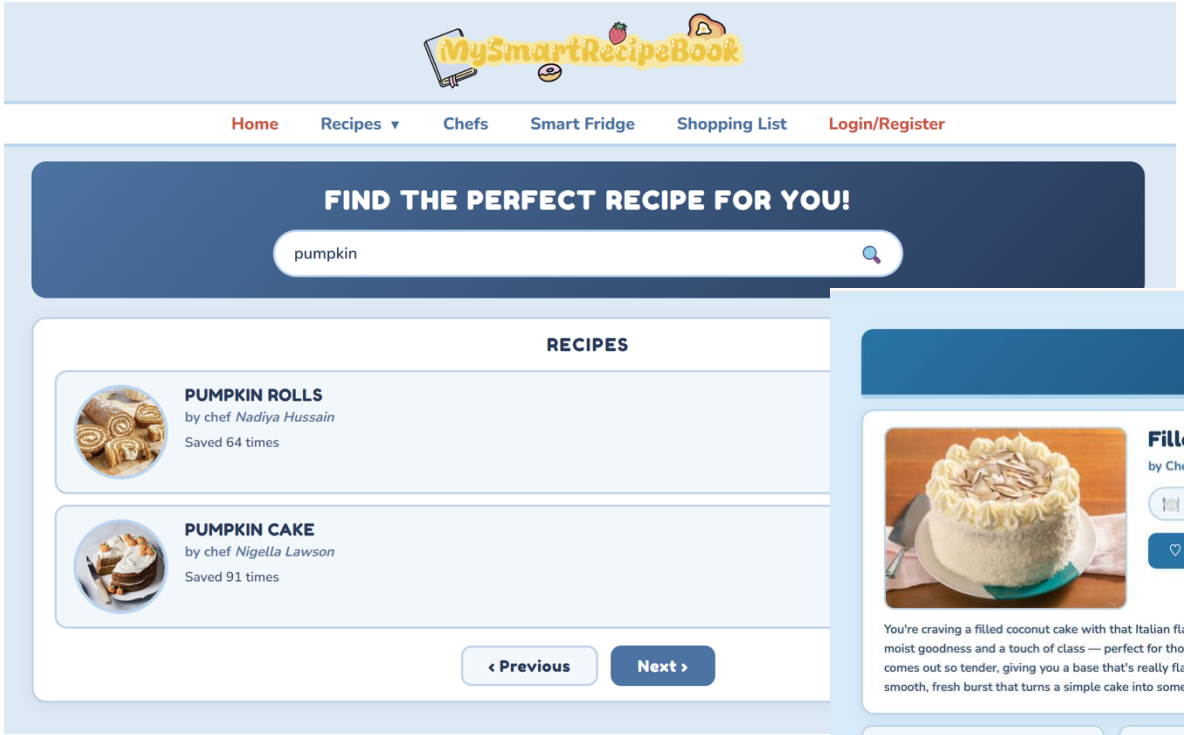
Unregistered User

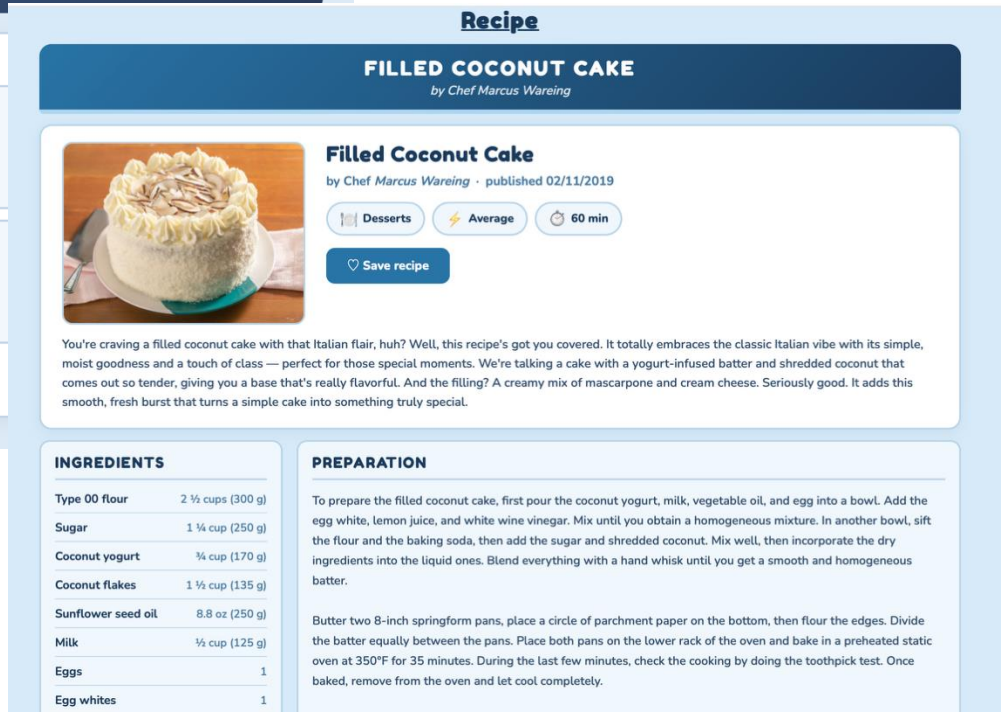


Unregistered users can browse chefs, searching them by surname and explore their published recipes.

Actors and Mockups (v)

Unregistered User

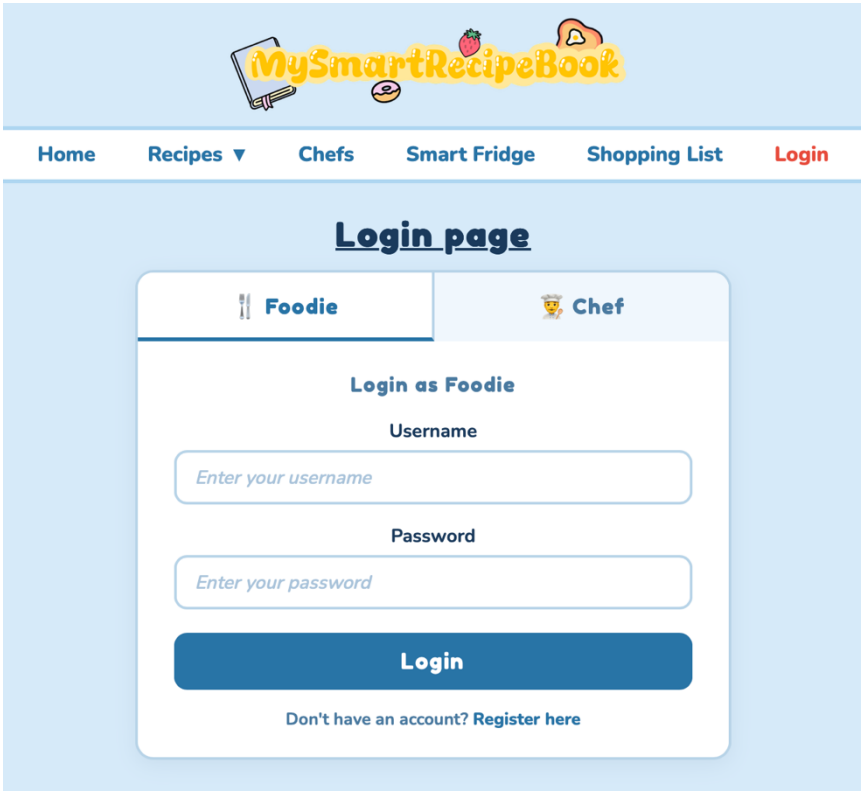


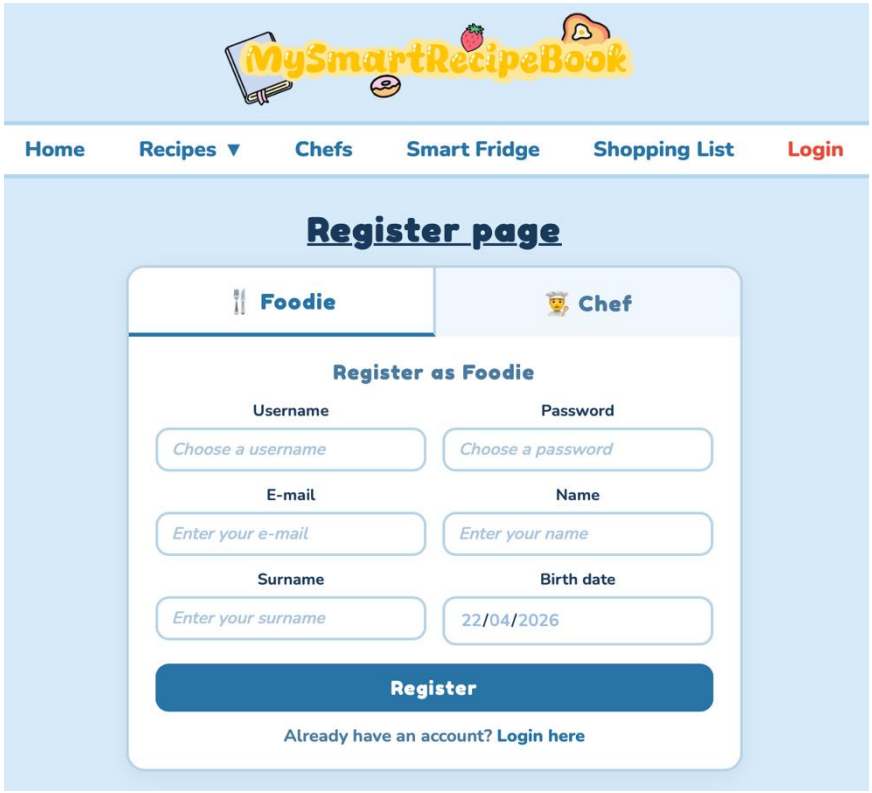


Unregistered users can search recipes by keyword, browse matching results, and access the complete recipe view with all the detailed information.

Actors and Mockups (vi)

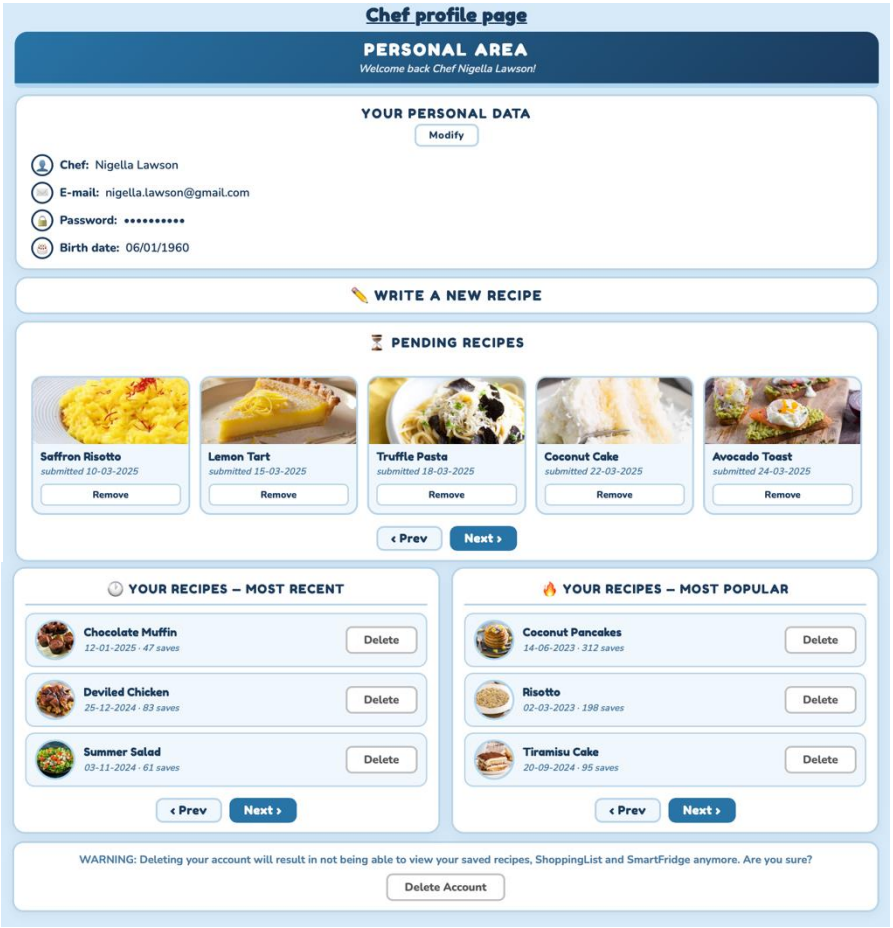
Log In and Register

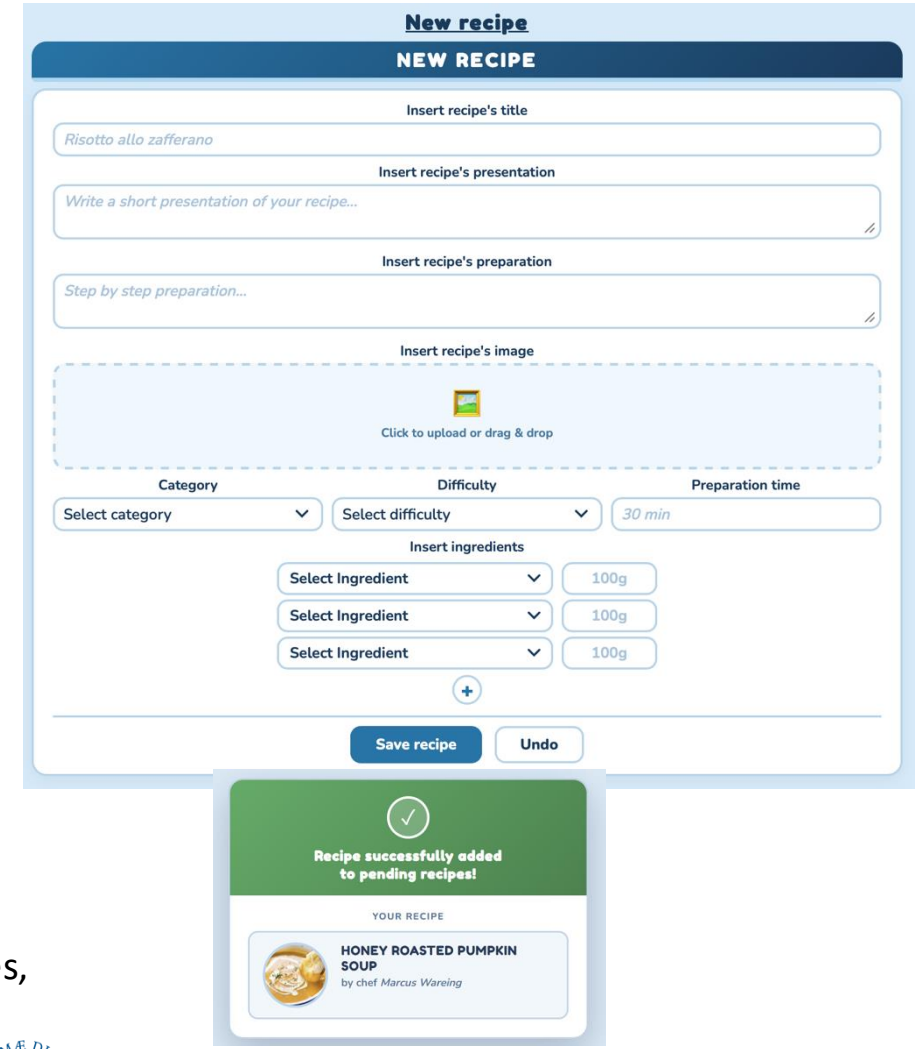




Actors and Mockups (vii)

Chef





The Chef can manage personal info, create new recipes, monitor pending submissions and see their recipes.

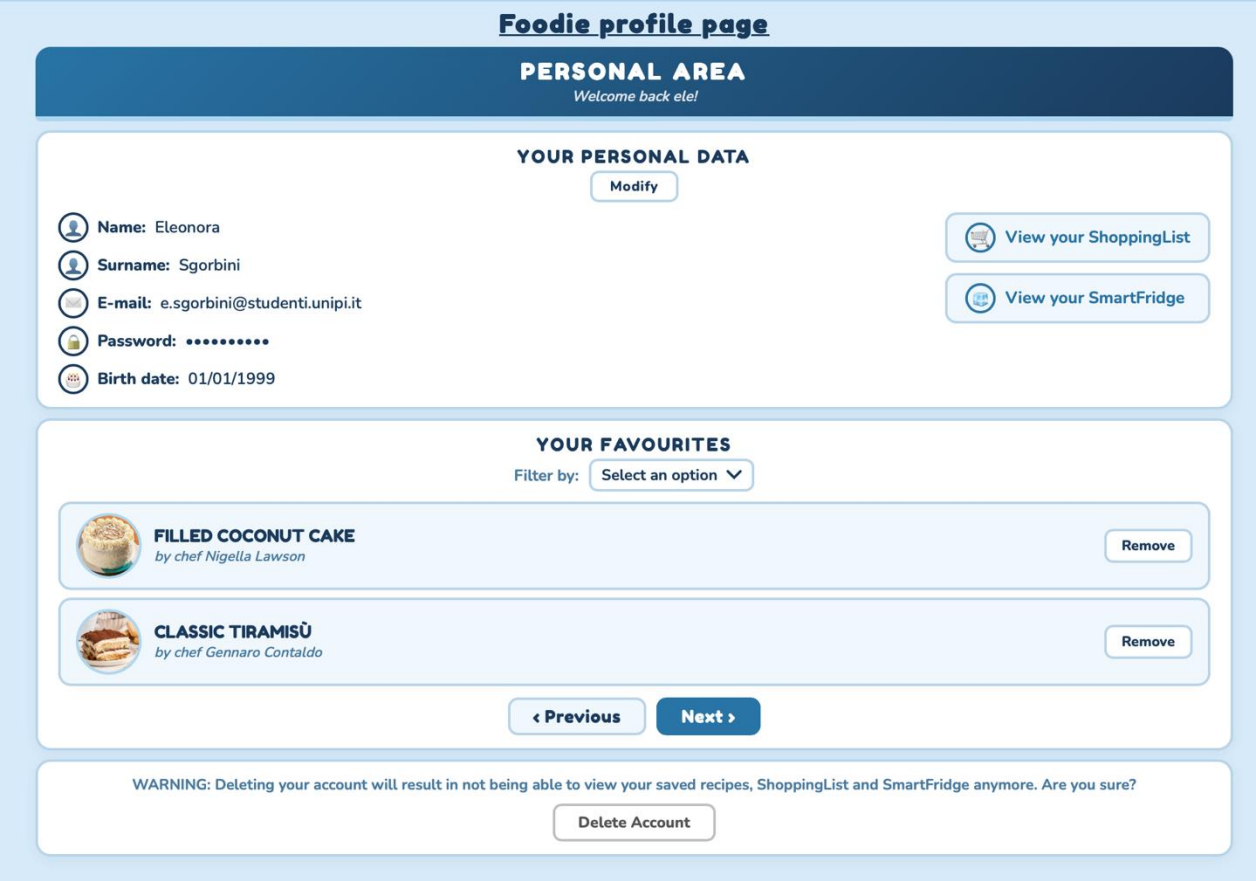
Actors and Mockups (viii)

Admin

The Admin can handle pending recipes , chef registration requests and view the analytics by clicking the corresponding button.

Actors and Mockups (ix)

Foodie

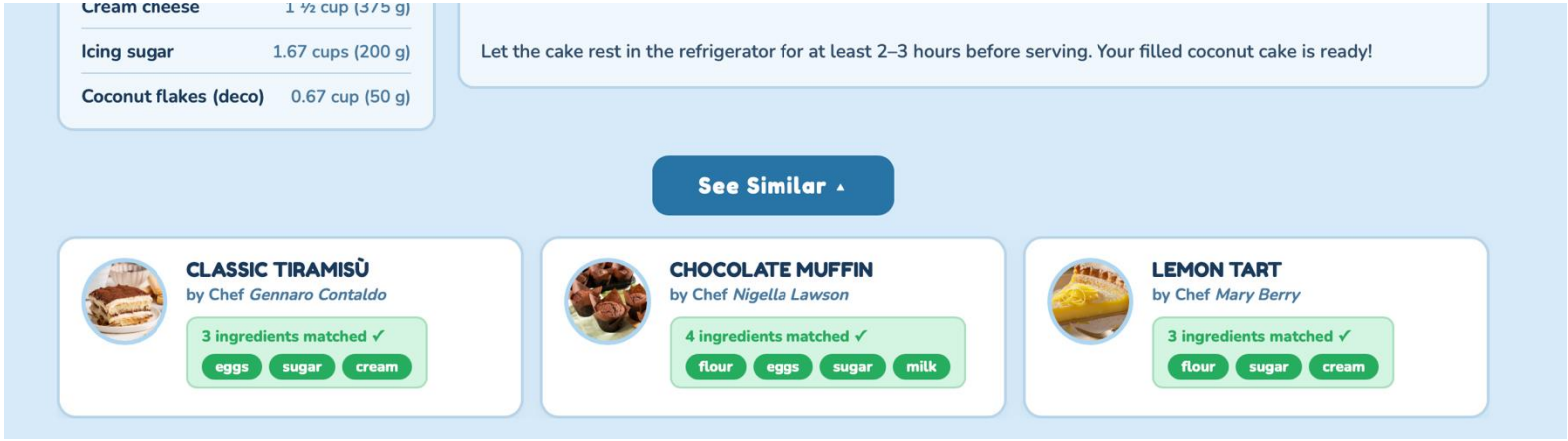


The Foodie manages personal information, accesses Smart Fridge and Shopping List features, and keeps track of favourite recipes.

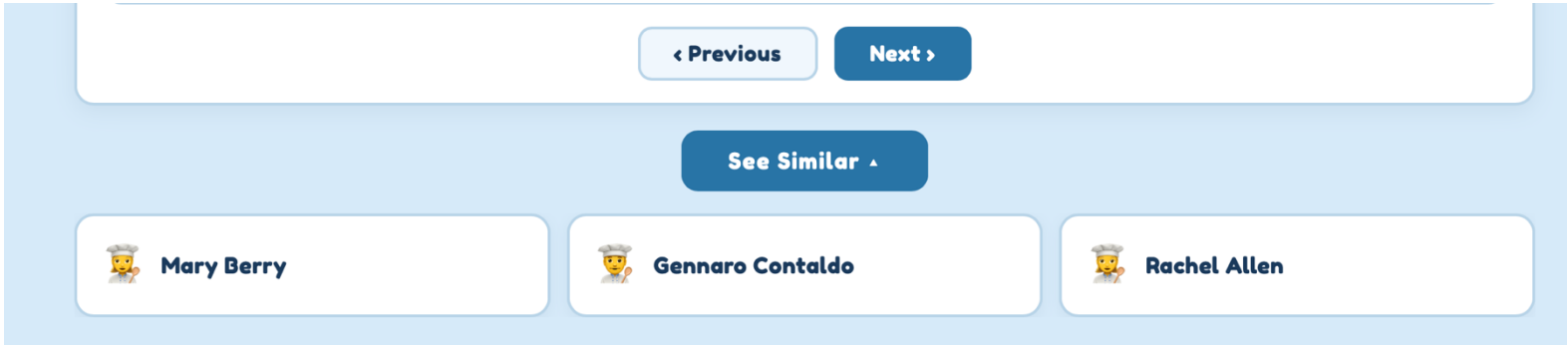
Actors and Mockups (x)

Foodie – See Similar

Recipes

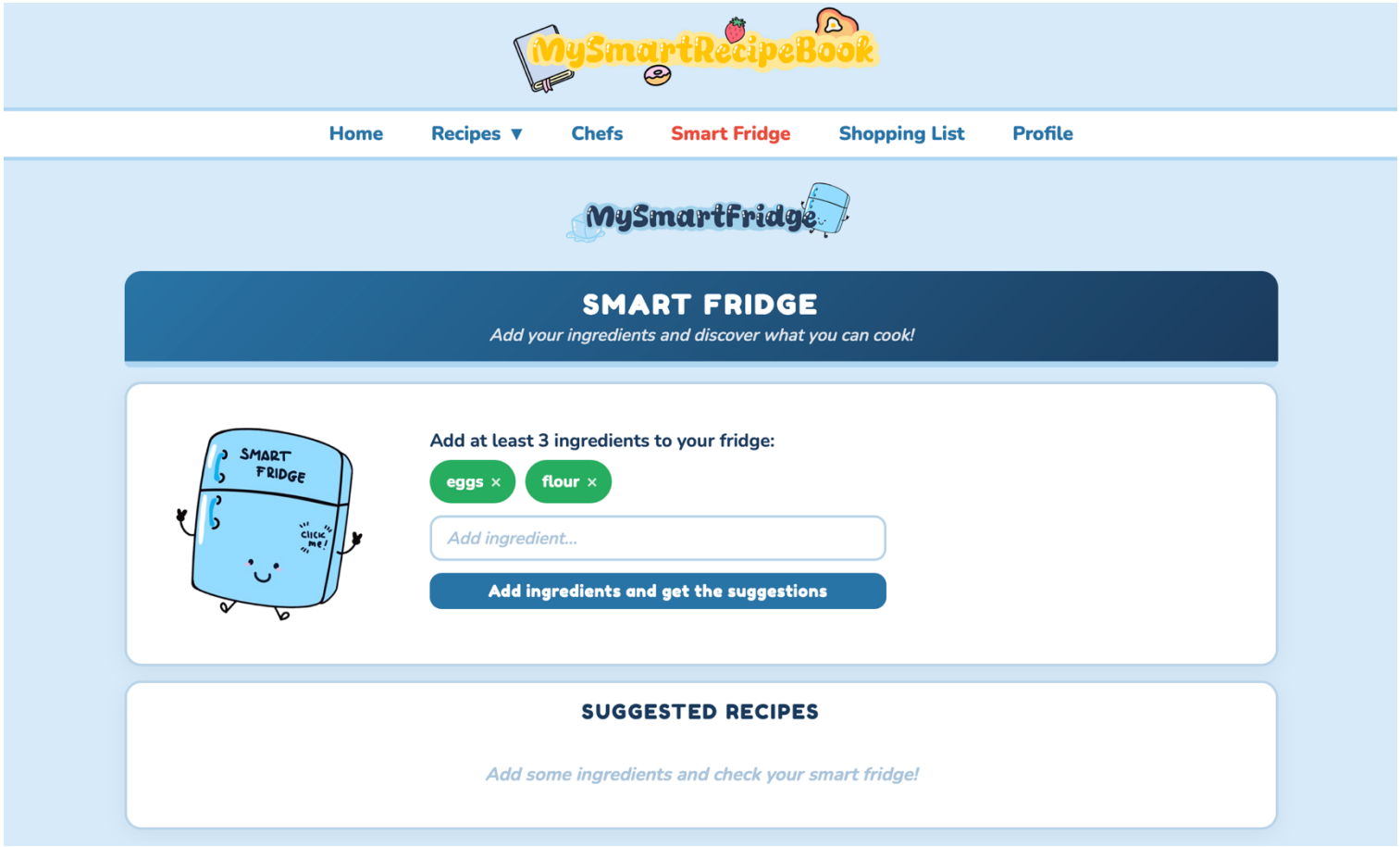


Chefs



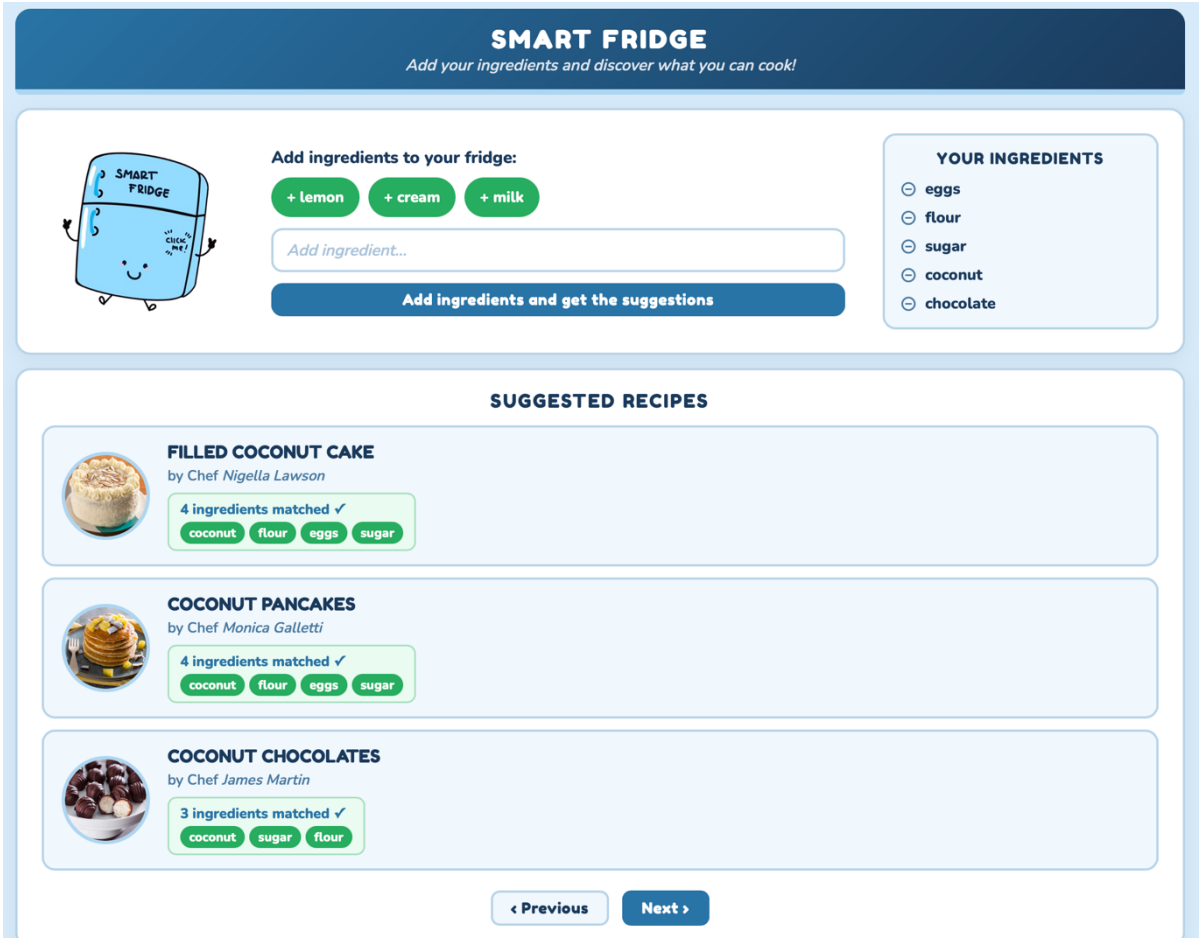
Actors and Mockups (xi)

Foodie – Smart Fridge



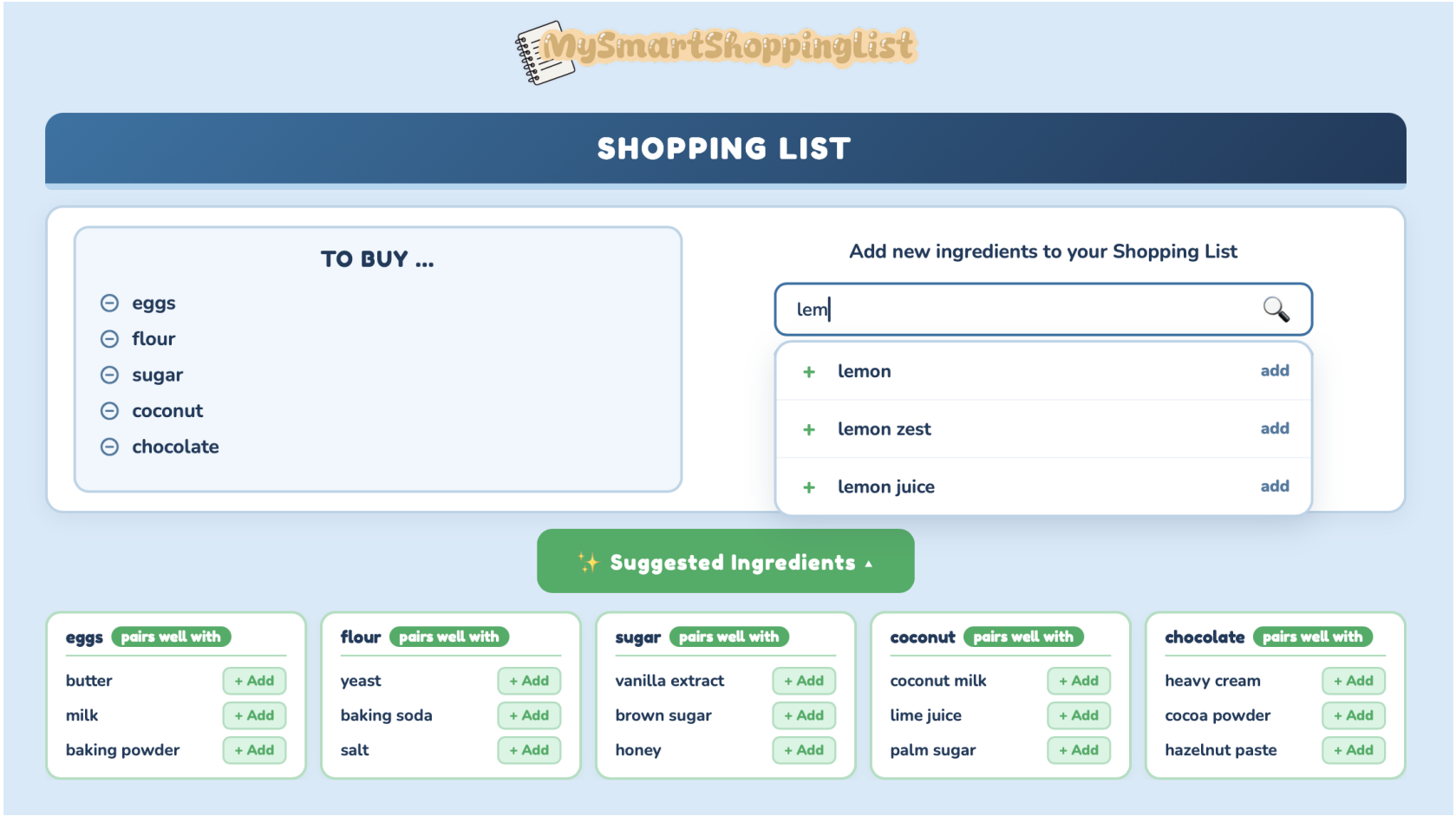
Actors and Mockups (xii)

Foodie – Smart Fridge



Actors and Mockups (xiii)

Foodie – Smart Shopping List



Dataset Description

Sources (via Python Web-Scraping):

- BBC GoodFood
- AllRecipes
- GialloZafferano

Volume:

- 15,832 Recipes — ≈ 53 MB
- 5,000 Foodies
- 93 Chefs

} ≈ 2.5 MB

Variety:

It is achieved by the fact that the web-service handles data retrieved from three different sources.

Velocity/Variability:

The system is highly dynamic, evolving rapidly through continuous data ingestion and user activity. This includes the regular insertion of new recipes, ongoing user registrations, recipe saving actions, dynamic Smart Fridge interactions, and real-time Shopping List updates.

Dataset Description

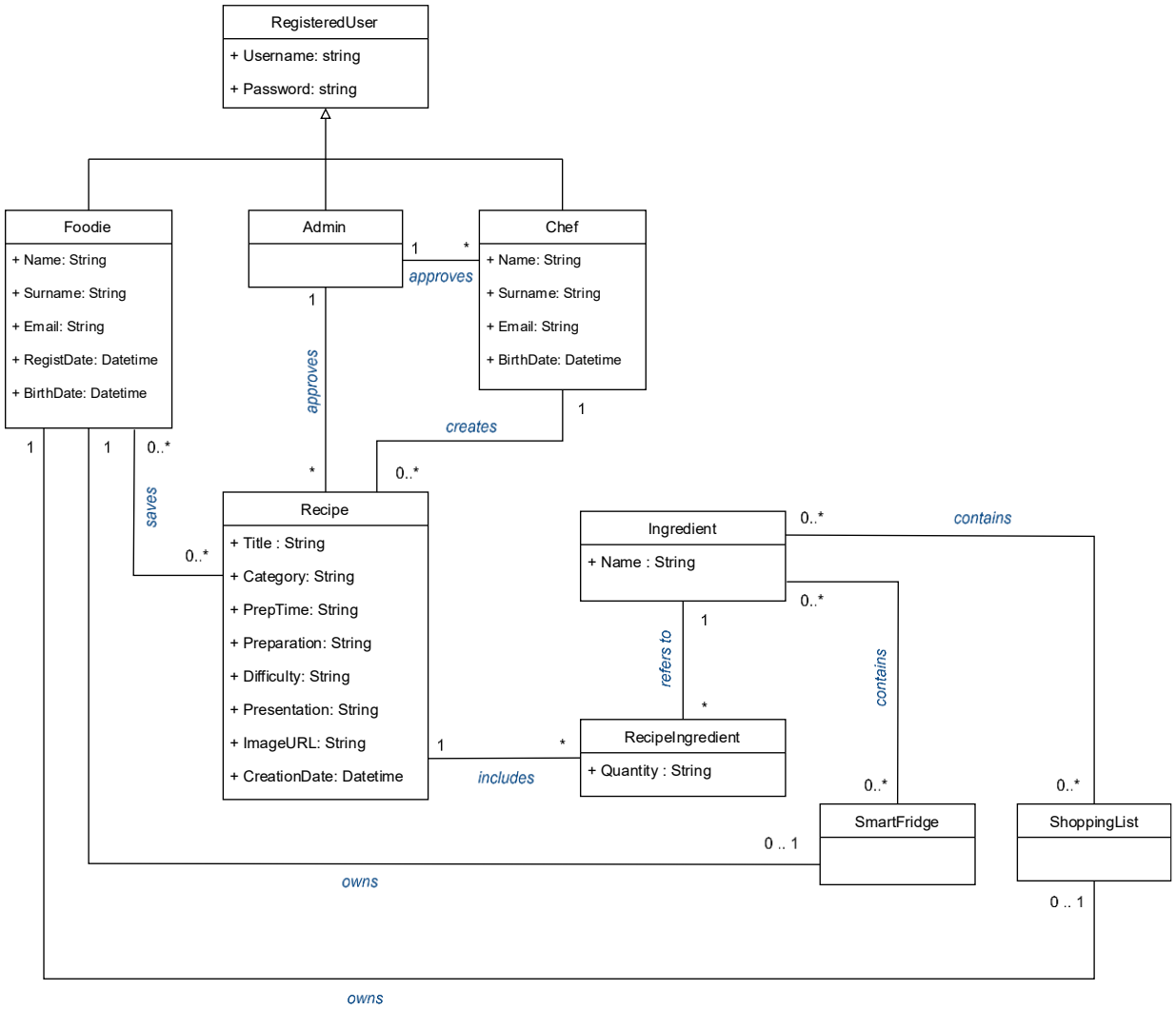
Recipes

- **Web Scraping Sources**
- **Deduplication:** Removed duplicate recipes.
- **Ingredient Normalization:** Normalized ingredients for consistent data within the application.
- **Text Cleaning:** Eliminated artifacts and redundant formatting characters.
- **Missing Data Generation:** Manually integrated or deduced missing attributes by re-consulting sources and cross-referencing web data.

Users

- **Foodie Profiles :** Generated using the Python *Faker* library to populate realistic personal attributes (names, emails, registration dates, birthdays).
- **Chef Profiles:** Real chefs were scraped from culinary websites to ensure authentic professional identities, while all other metadata was generated via Python *Faker* library.

UML Design Class Diagram



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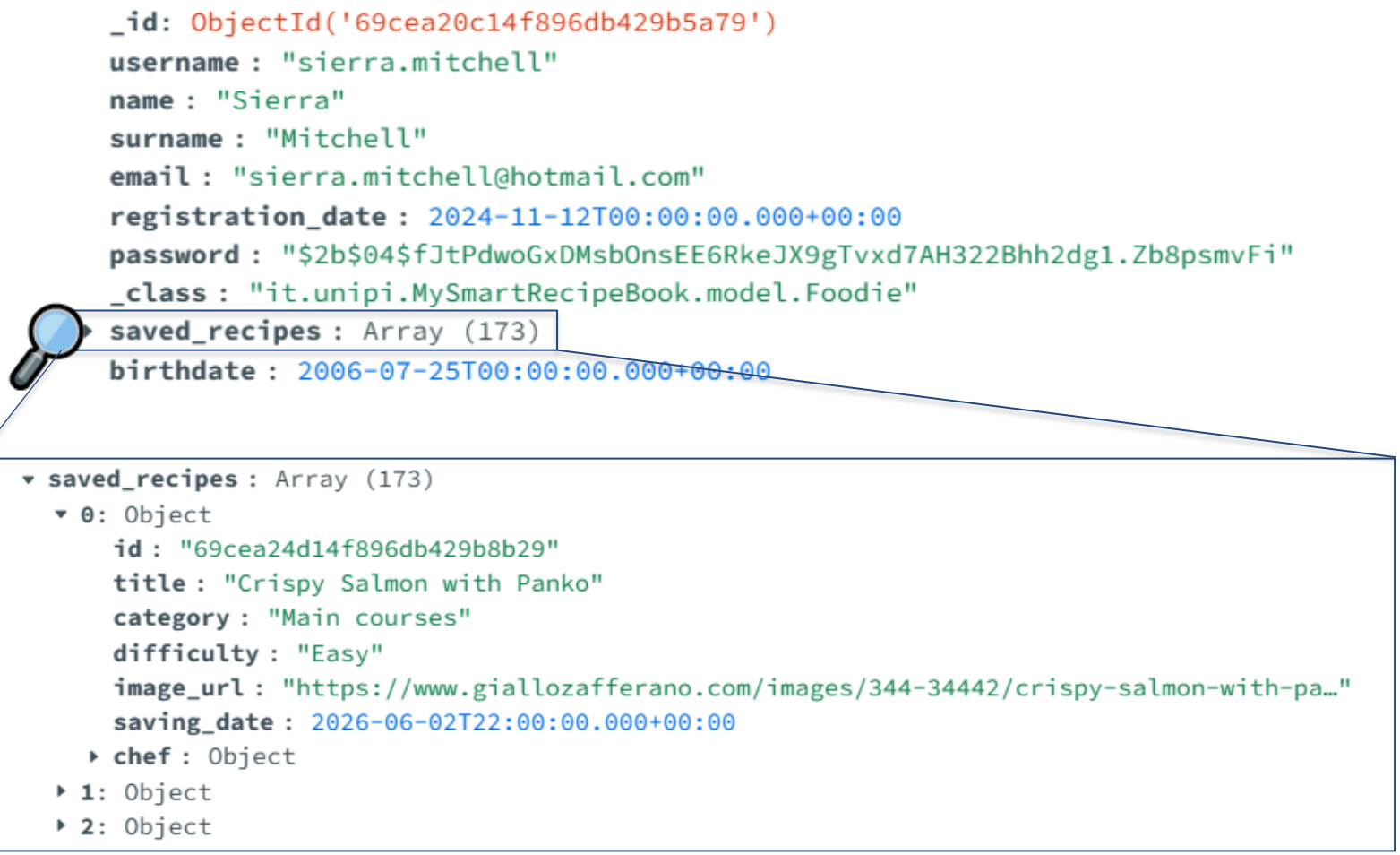
Application non-functional requirements

The **non functional-requirements** of our system are the following:

- The system must encrypt the user's password.
- The system must be developed by using an OOP language (Java).
- The system must be a responsive website application.
- The system must be able to handle high traffic.
- The system must implement a multi-database architecture using DocumentDB, GraphDB, and Key-Value DB.
- The system must expose all its functionalities by means of well-documented RESTful APIs, in accordance with the OpenAPI specification.

Document DB Design

Foodie



Document DB Design

Chef

```
_id: ObjectId('69ce825a14f896db429a146f')
username : "nadiya.hussain"
new_recipes : Array (15)
old_recipes : Array (147)
popular_recipes : Array (16)
tot_saves : 5206
tot_recipes : 162
status : "APPROVED"
name : "Nadiya"
surname : "Hussain"
email : "nadiya.hussain@yahoo.com"
password : "$2b$04$178iTz6zwQsFYR7r1eSuIu0rEqJZx9manOjMx7cR5lUzkfsGE.X/a"
birthdate : 1949-05-18T23:00:00.000+00:00
_class : "it.unipi.MySmartRecipeBook.model.Mongo.users.Chef"
recipes_to_confirm : Array (1)
  0: Object
      id : "6a2180330fc2e5349e11dc84"
      title : "Sweet Cherry Pie"
      image_url : "https://images.unsplash.com/photow=800&q=80"
      creation_date : 2026-06-04T13:40:03.537+00:00
```

Document DB Design

Chef

Document DB Design

Admin


```
_id: ObjectId('69cea60d14f896db429babe3')
_class : "it.unipi.MySmartRecipeBook.model.Admin"
password : "$2a$10$ZWl8yOftWkI36aJQbvLHh.b.BqYYEVPH6tvHHfgmBzMiSbGma1jWC"
username : "admin"
▾ recipes_to_approve : Array (1)
  ▾ 0: Object
    ▾ chef : Object
        id : "69ce825a14f896db429a146d"
        name : "Nigella"
        surname : "Lawson"
      id : "6a2a7b133f6583321a0ffd1e"
      title : "Sweet Apple Pie"
      image_url : "https://images.unsplash.com/photow=800&q=80"
      creation_date : 2026-06-11T09:08:35.704+00:00
▾ chefs_to_approve : Array (1)
  ▾ 0: Object
      _id : ObjectId('6a2a7d783f6583321a0ffd1f')
      username : "garrett.graham"
      name : "Garrett"
      surname : "Graham"
```

Document DB Design

Recipe


```
_id: ObjectId('69cea24b14f896db429b6e0f')
title : "Easy Pineapple Upside-Down Cake"
image_url : "https://www.allrecipes.com/thmb/HV7Yth9XMlJD4nur5MsQwlnbdLA=/1500x0/fi…"
category : "Desserts"
difficulty : "Easy"
prep_time : "15 mins"
presentation : "Here's a quick and easy pineapple upside-down cake recipe using any br…"
▾ ingredients : Array (9)
  ▸ 0: Object
  ▸ 1: Object
  ▸ 2: Object
  ▾ 3: Object
      name : "pecans"
      quantity : "2 tablespoons"
  ▸ 4: Object
  ▸ 5: Object
  ▸ 6: Object
  ▸ 7: Object
  ▸ 8: Object
preparation : "Preheat the oven to 350 degrees F (175 degrees C).  Melt butter in a 9…"
▾ chef : Object
    id : "69ce825a14f896db429a146f"
    name : "Nadiya"
    surname : "Hussain"
creation_date : "2021-05-12T19:43:39.124"
_class : "it.unipi.MySmartRecipeBook.model.Mongo.RecipeMongo"
num_saves : 29
status : "APPROVED"
```

Relevant MongoDB queries

Chef Ranking using Bayesian Score

```
@Aggregation(pipeline = {
        "{ $match: { username: { $ne: 'admin' }, status: { $ne: 'pending' } } }",

        "{ $addFields: { " +
                "   tot_saves: { $ifNull: ['$tot_saves', 0] }, " +
                "   tot_recipes: { $ifNull: ['$tot_recipes', 0] } " +
                "} }",

        "{ $setWindowFields: { partitionBy: null, output: { " +
                "totalSavesGlobal: { $sum: '$tot_saves' }, " +
                "totalRecipesGlobal: { $sum: '$tot_recipes' } } } }",

        "{ $addFields: { C: { $cond: [ { $eq: ['$totalRecipesGlobal', 0] }, 0, " +
                "{ $divide: ['$totalSavesGlobal', '$totalRecipesGlobal'] } ] } } }",

        "{ $addFields: { R: { $cond: [ { $eq: ['$tot_recipes', 0] }, 0, " +
                "{ $divide: ['$tot_saves', '$tot_recipes'] } ] } } }",

        "{ $addFields: { score: { $add: [ " +
                "{ $multiply: [ { $divide: ['$tot_recipes', { $add: ['$tot_recipes', 5] }] }, '$R' ] }, " +
                "{ $multiply: [ { $divide: [5, { $add: ['$tot_recipes', 5] }] }, '$C' ] } " +
                "] } } }",

        "{ $sort: { score: -1 } }",

        "{ $setWindowFields: { sortBy: { score: -1 }, output: { rank: { $rank: {} } } } }",
        "{ $limit: 5 }",
        "{ $project: { rank: 1, name: 1, surname: 1, score: 1 } }"
})
List<ChefRankAnalyticsDTO> chefBayesianRanking();
```

This query ranks chefs through a Bayesian score that combines their total number of saves and the total number of recipes, highlighting the most consistently appreciated profiles.

Relevant MongoDB queries

Monthly Foodie Registration Statistics

```
@Aggregation(pipeline = {
        "{ $match: { $expr: { $eq: [{ $year: '$registration_date' }, ?0] } } }",
        "{ $group: { " +
                "         _id: { $dateToString: { format: '%Y-%m', date: '$registration_date' } }, " +
                "         year: { $first: { $dateToString: { format: '%Y', date: '$registration_date' } } }, " +
                "         number: { $sum: 1 } " +
                "} }",

        "{ $sort: { 'year': -1, 'number': -1 } }",

        "{ $group: { " +
                "         _id: '$year', " +
                "         totalRegisteredFoodies: { $sum: '$number' }, " +
                "         monthAnalyticsDTOList: { $push: { month: '$_id', totalFoodies: '$number' } } " +
                "} }",

        "{ $project: { " +
                "         _id: 0, " +
                "         year: { $toInt: '$_id' }, " +
                "         totalRegisteredFoodies: 1, " +
                "         monthAnalyticsDTOList: 1 " +
                "} }"
})
YearAnalyticsDTO getMonthlyFoodiesStats(int year);
```

Filters Foodie registrations by a target year to compute the overall annual growth and detailed monthly acquisition metrics

Relevant MongoDB queries

Category Trend Analysis

```
@Aggregation(pipeline = {
        "{ $match: { status: { $ne: 'pending' } } }",
        "{ $addFields: { " +
                "is_recent: { $gte: [ { $toDate: '$creation_date' }, ?0 ] }, " +
                "is_previous: { $and: [ " +
                "{ $lt: [ { $toDate: '$creation_date' }, ?0 ] }, " +
                "{ $gte: [ { $toDate: '$creation_date' }, ?1 ] } " +
                "] } " +
                "} }",
        "{ $group: { " +
                "_id: '$category', " +
                "recentCount: { $sum: { $cond: [ '$is_recent', 1, 0 ] } }, " +
                "previousCount: { $sum: { $cond: [ '$is_previous', 1, 0 ] } } " +
                "} }",

        "{ $addFields: { " +
                "growthRate: { $cond: [ " +
                "{ $gt: [ '$previousCount', 0 ] }, " +
                "{ $divide: [ { $subtract: [ '$recentCount', '$previousCount' ] }, '$previousCount' ] }, " +
                "null " +
                "] } " +
                "} }"
})
List<TrendAnalyticsDTO> findCategoryTrend(LocalDateTime recentDate, LocalDateTime previousDate);
```

Compares the number of recipes created in a recent period with a previous one, identifying which categories are growing or decreasing over time.

Relevant MongoDB queries

Most Saved Recipes per Category

```
@Aggregation(pipeline = {
        "{ $match: { status: 'APPROVED' } }",
        "{ $sort: { num_saves: -1 } }",
        "{ $group: { " +
                "   _id: '$category', " +
                "   recipeId: { $first: '$_id' }, " +
                "   title: { $first: '$title' }, " +
                "   chef: { $first: '$chef' }, " +
                "   num_saves: { $first: '$num_saves' }, " +
                "   image_url: { $first: '$image_url' }" +
                "} }",
        "{ $project: { " +
                "   _id: '$recipeId', " +
                "   category: '$_id', " +
                "   title: 1, " +
                "   chef: 1, " +
                "   num_saves: 1, " +
                "   image_url: 1" +
                "} }"
})
List<RecipeMongo> findMostSavedRecipePerCategory();
```

This query identifies the most saved approved recipe for each category, highlighting the recipes that users appreciate the most within every food category.

MongoDB Replica Set Configuration

Three replicas were implemented with the following configuration:

```
{ _id: 0, host: 'mongo_machine1:10.1.1.92', priority: 5 }
{ _id: 1, host: 'mongo_machine2:10.1.1.91', priority: 2 }
{ _id: 2, host: 'mongo_machine3:10.1.1.90', priority: 1 }
```

The priority parameters are chosen to handle the election process in the event of a primary node failure.

Write Concern and Read Preference

- **Write Concern: Write concern 1 (W1).** This ensures fast write operations and low latency, accepting a minimal risk of data loss in the rare event of a Primary failure immediately before the asynchronous replication to Secondaries occurs.
- **Read Preference: Nearest**. This effectively distributes the read load across the available infrastructure and minimizes query response times.

MongoDB Indexes

Recipes Collection

`creation_date` (Descending Index)

Descending index on the creation_date field was added to drastically optimize the retrieval of the newest recipes

Metric	Without index	With index
Documents returned	5	5
Execution time	73 ms	2 ms
Index keys examined	0	5
Documents examined	15,832	5

`title` (Text Index)

Users search via partial keywords or multi-word phrases (findByTitleContainingIgnoreCase).

Metric	Without index	With Text index
Documents returned	28	28
Execution time	58 ms	22 ms
Index keys examined	0	1010
Documents examined	15,832	1060

`category` (Standard Index)

Added to optimize category-based recipe browsing, one of the most frequent operation in the application.

Metric	Without index	With index
Documents returned	3011	3011
Execution time	96 ms	19 ms
Index keys examined	0	3011
Documents examined	15,832	3011

MongoDB Indexes

Chefs Collection

`username` (Unique Index)

Optimizes login, registration and profile retrieval and garantees the uniqueness of the usernames.

Metric	Without index	With index
Documents returned	1	1
Execution time	0 ms	0 ms
Index keys examined	0	1
Documents examined	94	1

`surname` (Standard Index) ✗ Not retained

Evaluated for improving surname-based chef search. Considered not strictly necessary in the current version, it could become relevant as the collection grows.

Foodies Collection

`username` (Unique Index)

Introduced for data integrity during registration.

- Ensures no two foodies can register with the same username.
- Useful for the most frequent operations such as login

Discussion on MongoDB Data Sharding

MongoDB

- **Foodies (Hashed on** username**):** Uniform data distribution. Optimizes the authentication process by enabling highly efficient, direct lookups without broadcast queries.
- **Chefs (Future: Hashed on** _id**):** Currently unsharded due to low volume. The future strategy prioritizes high-frequency profile reads over a slight overhead during login.
- **Recipes (Hashed on** _id**):** Optimizes the retrieval of recipe details (dominant operation).

Redis (Smart Fridge & Shopping List)

- **Current Architecture:** Sentinel Master-Slave. The workload is heavily read-bound, making a Cluster/Sharding approach unnecessary at this stage.
- **Future Development:** If write operations increase, the system will migrate to a Redis Cluster. It will adopt a key hashing strategy using *Hash Tags* to ensure all data for a single user is routed to the same shard.

Key-value DB Design

Keys Naming Conventions across the different features

Shopping List

```
Foodie: <username> : shoppingList
```

Smart Fridge

```
Foodie: <username> : smartFridge : ingredients
Foodie: <username> : smartFridge : suggestions
```

Allowed Ingredients

```
Allowed_ingredients
```

Key-value DB Design

Allowed Ingredients

Centralized in-memory set that validates ingredients across the whole platform, both in Foodies' lists and Chefs' recipe creation.

```
SMEMBERS Allowed_ingredients
    "pork belly"
    "mascarpone cheese"
    "brown sugar"
    "..."
```

Shopping List

Set of ingredients a Foodie plans to buy. Every input is normalized to lowercase before storage.

```
SMEMBERS Foodie:sierra.mitchell:shoppingList
    "butter"
    "bread"
    "spaghetti pasta"
    "avocados"
```

Key-value DB Design

Smart Fridge – Ingredients

Set of ingredients a Foodie has at home (same lowercase normalization as the Shopping List). A recipe is suggested when at least 3 of them match.

```
SMEMBERS
Foodie:sierra.mitchell:smartFridge:ingredients
     ”bread"
     "butter"
     "avocados“
```

Smart Fridge – Suggestions

Computed via the Graph DB, then cached in Redis as a sorted JSON list to reduce latency.

```
GET Foodie:sierra.mitchell:smartFridge:suggestions
  [ {
      "id" : "69cea24e14f896db429ba7b7",
      "title" : "Best Chicken Panini",
      "chef" : "Gordon Ramsay",
      "matchCount" : 3,
      "matchedIngredients" : [ "bread", "butter",
      "avocados" ]
  }, { ... } ]
```

Redis Replica Set Configuration

To guarantee high availability and eliminate a critical Single Point of Failure (SPOF) for caching and session management, a **Master-Replica** architecture was implemented.

The data topology distributes:

- Single **Master** node on machine 10.1.1.92
- Two **Replica** nodes on machines 10.1.1.91 and 10.1.1.90

All nodes are operating on port 6379 and Sentinel on port 7004.

To ensure fault tolerance, **Redis Sentinel** is configured across the infrastructure, in order to constantly monitor the state of the nodes.

Relevant Redis queries

Ingredients Management

```
public Set<String> getAllAllowedIngredients() {
    try (Jedis jedis = jedisSentinelPool.getResource()) {
        return jedis.smembers(INGREDIENTS_REDIS_KEY);
    }
    catch (JedisException e) {
        return new HashSet<>();
    }
}
```

Shopping List & Smart Fridge Management – add ingredient(s)

```
String key = REDIS_KEY_PREFIX + foodie.getUsername() + REDIS_KEY_FRIDGE_SUFFIX;

if (!ingredients.isEmpty()) {
    try (Jedis jedis = jedisSentinelPool.getResource()) {
        jedis.sadd(key, ingredients.toArray(new String[0]));
        jedis.del( key: REDIS_KEY_PREFIX + foodie.getUsername() + REDIS_KEY_RECIPES_SUFFIX);
        jedis.expire(key,  seconds: 86400*15);

    }
```

Relevant Redis queries

Shopping List & Smart Fridge Management – remove ingredient

```
if(ingredientService.isValidIngredient(ingredient)) {
    String key = REDIS_KEY_PREFIX + foodie.getUsername() + REDIS_KEY_FRIDGE_SUFFIX;
    try (Jedis jedis = jedisSentinelPool.getResource()) {
        jedis.srem(key, ingredient);
        jedis.expire(key, seconds: 86400*15);
    }
    updateCacheAfterRemoval(foodie.getUsername(), ingredient);
}
```

Recipe Suggestions Caching

```
String cacheKey = REDIS_KEY_PREFIX + username + REDIS_KEY_RECIPES_SUFFIX;

String json;
try (Jedis jedis = jedisSentinelPool.getResource()) {
    json = jedis.get(cacheKey);
}
```

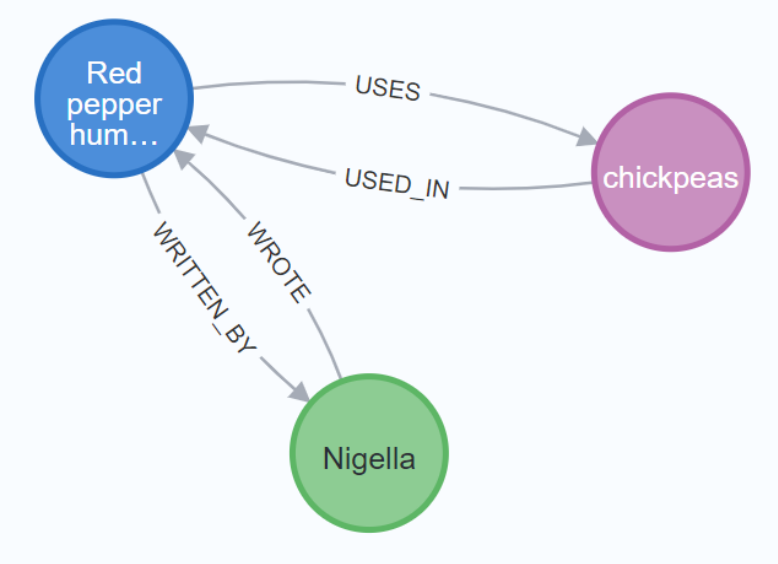
Graph DB Design

Nodes

Chef: authors of the recipes with basic information: *mongo_id*, name and surname.

Ingredient: ingredients used in recipes.

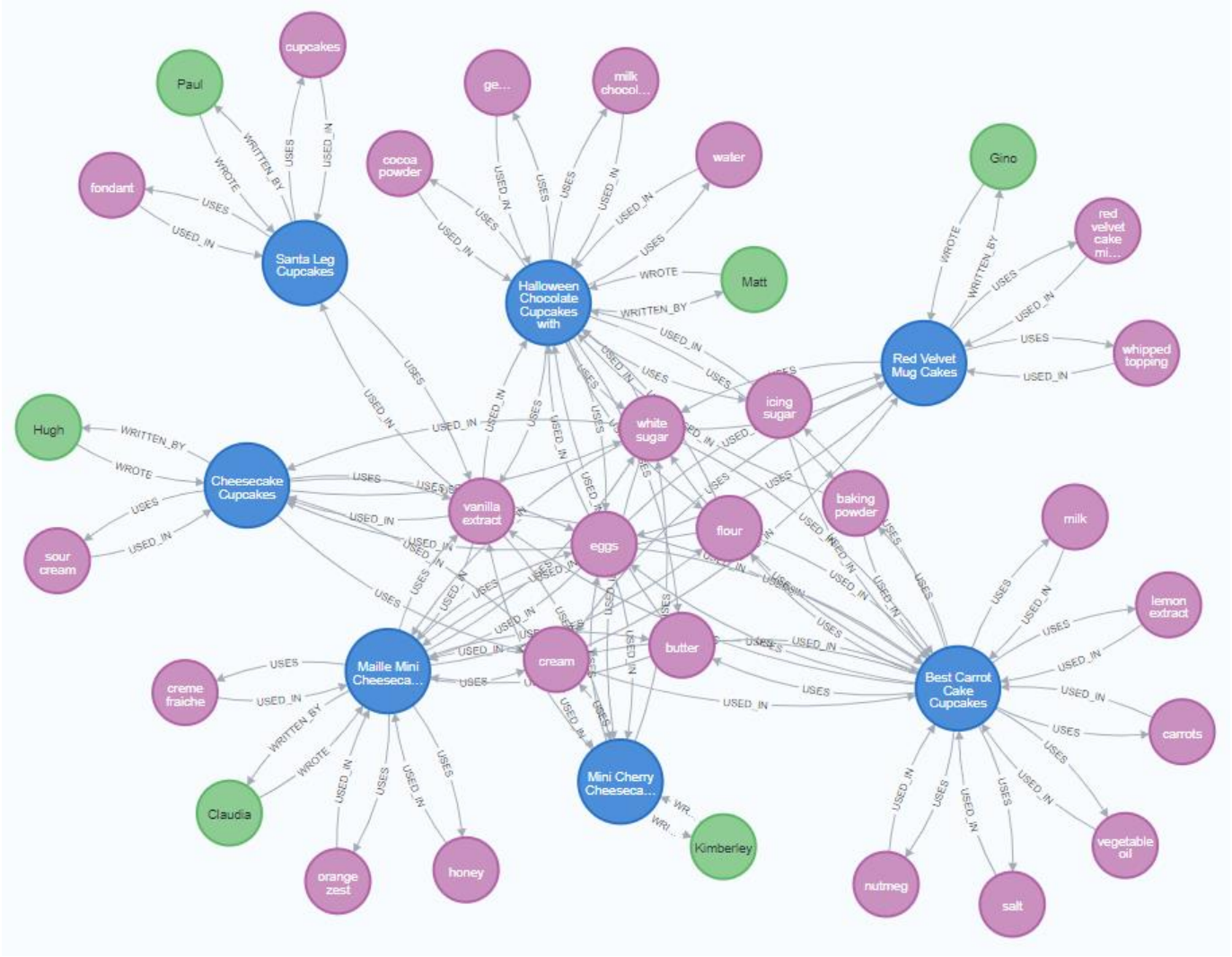
Recipe: recipes in the system. Each node contains: *mongo_id*, title, image URL and category.



Relations

- **:USED_IN**: connects an *Ingredient* to a *Recipe*, representing which ingredients are used in a given recipe.
- **:USES**: connects a recipe to an ingredient, is the inverse of USED_IN
- **:WRITTEN_BY**: connects a *Recipe* to a *Chef*, representing the author of the recipe.
- **:WROTE**: connects a *Chef* to a *Recipe*. Is the inverse navigation of *:WRITTEN_BY*

Graph DB Design



Relevant Neo4j queries

Similar Chefs by Shared Ingredients

```
@Query("MATCH (target:Chef {mongo_id: $targetMongoId})-[:WROTE]->(r1:Recipe)-[:USES]->(i:Ingredient) " +
        "WITH target, i, collect(DISTINCT r1.category) AS targetCategories " +
        "WHERE COUNT { (i)-[:USED_IN]->() } < 350 " +
        "WITH DISTINCT target, i, targetCategories " +
        "MATCH (i)-[:USED_IN]->(r2:Recipe)-[:WRITTEN_BY]->(other:Chef) " +
        "WHERE target <> other AND r2.category IN targetCategories " +
        "WITH other, count(DISTINCT i) AS sharedIngredientsScore " +
        "ORDER BY sharedIngredientsScore DESC " +
        "LIMIT 3 " +
        "RETURN other.mongo_id AS id, " +
        "       other.name AS name, " +
        "       other.surname AS surname")
List<ChefInfoDTO> findSimilarChefs(@Param("targetMongoId") String targetMong
```

Identifies most similar chefs by analyzing shared specialized ingredients within matching recipe categories.

Relevant Neo4j queries

Suggested Complementary Ingredients

```
@Query("MATCH (target:Ingredient)-[:USED_IN]->(r:Recipe)-[:USES]->(other:Ingredient) " +
        "WHERE target.name IN $ingredientList " +
        "AND NOT other.name IN $ignoredIngredients " +
        "WITH target.name AS originalIngredient, other.name AS suggestedIngredient, count(r) AS occurrences " +
        "ORDER BY occurrences DESC " +
        "RETURN originalIngredient, collect(suggestedIngredient)[0..3] AS suggestedIngredients")
List<IngredientSuggestionDTO> findSuggestedIngredientsForList(
        @Param("ingredientList") List<String> ingredientList,
        @Param("ignoredIngredients") List<String> ignoredIngredients
);
```

Recommends complementary ingredients by finding the ones most frequently paired with the user's current shopping list, filtering out the list itself and most common ingredients

Relevant Neo4j queries

Similar Recipes

```
@Query("MATCH (target:Recipe {mongo_id: $recipeId})-[:USES]->(i:Ingredient)-[:USED_IN]->(other:Recipe) " +
        "WHERE target <> other " +
        "WITH other, count(i) AS sharedIngredientsCount, collect(i.name) AS sharedIngredients " +
        "MATCH (other)-[:WRITTEN_BY]->(c:Chef) " +
        "RETURN other.mongo_id AS id, " +
        "       other.title AS title, " +
        "       other.imageURL AS imageURL, " +
        "       c.name AS chefName, " +
        "       c.surname AS chefSurname, " +
        "       c.mongo_id AS chefId, " +
        "       sharedIngredientsCount AS matchCount, " +
        "       sharedIngredients AS matchedIngredients " +
        "ORDER BY matchCount DESC, other.title ASC " +
        "LIMIT 3")
List<RecipeSuggestionDTO> findSimilarRecipes(String recipeId);
```

The query recommends similar recipes by exploiting ingredient-based relationships, ranking results according to the number of shared ingredients with the selected recipe.

Relevant Neo4j queries

Smart Fridge Recipe Suggestions

```
@Query("MATCH (i:Ingredient)-[:USED_IN]->(r:Recipe) " +
        "WHERE i.name IN $myIngredients " +
        "WITH r, count(i) AS matchCount, collect(i.name) AS matchedIngredients " +
        "WHERE matchCount >= 3 " +
        "MATCH (r)-[:WRITTEN_BY]->(c:Chef) " +
        "RETURN r.mongo_id AS id, " +
        "       r.title AS title, " +
        "       r.imageURL AS imageURL, " +
        "       c.name AS chefName, " +
        "       c.surname AS chefSurname, " +
        "       c.mongo_id AS chefId, " +
        "       matchCount, " +
        "       matchedIngredients " +
        "ORDER BY matchCount DESC")
List<RecipeSuggestionDTO> findRecipesByIngredients(List<String> myIngredients);
```

The query suggests recipes that uses at least three ingredients of the foodie's smart fridge, prioritizing those with the highest number of ingredient matches.

Neo4j Indexes

All Neo4j queries start from a specific node identified by a unique property. Without indexes, every lookup performs a full AllNodesScan (~18,859 nodes).

Performance Tests

- Execution of most relevant queries
- Tests repeated 5 times clearing query cache between each request

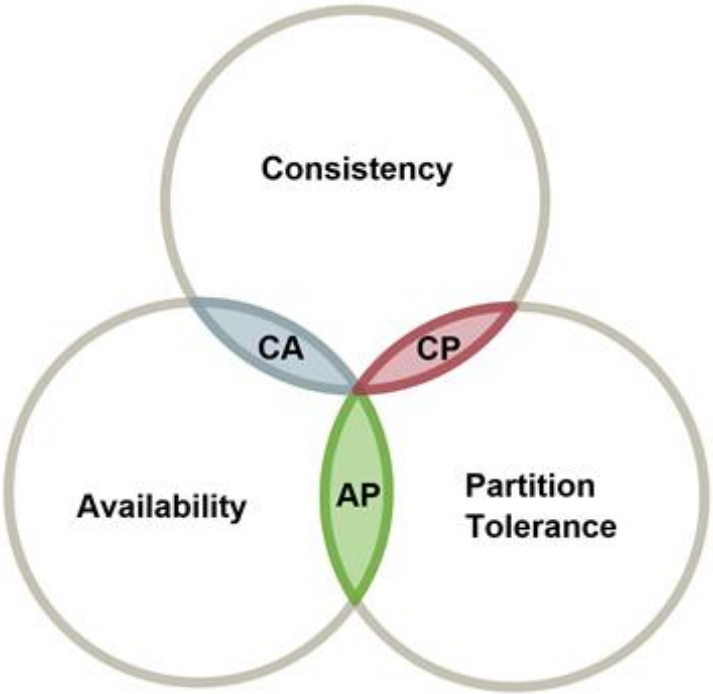
All indexes transform AllNodesScan (full graph scan) into NodeIndexSeek (direct access), eliminating the expensive initial lookup phase.

Query	Index	DB Hits (before → after)	Time
Smart Fridge Suggestion	ingredient_name_idx	51,733 → 11,080	53 ms → 28 ms
Find Similar Chef	chef_mongo_id_idx	661,062 → 623,252	206 ms → 191 ms
Similar Recipes Suggestion	recipe_mongo_idx	79,948 → 39,296	100 ms → 62 ms

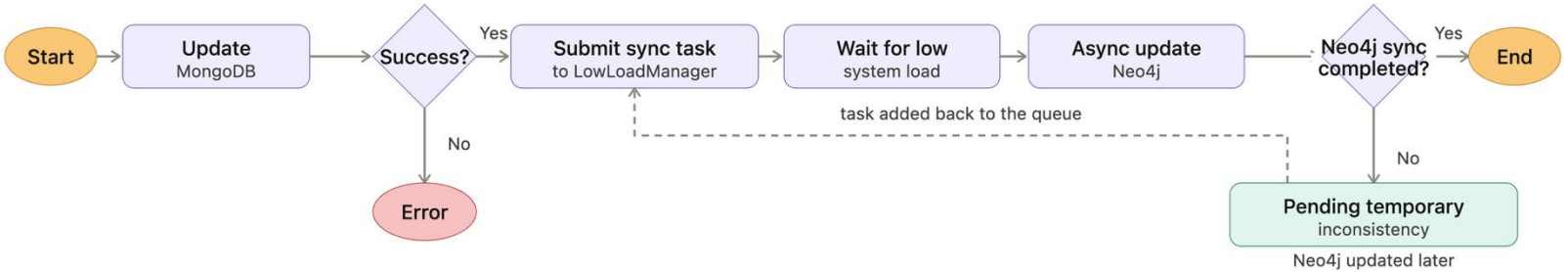
CAP – theorem consideration

The two selected properties from the CAP theorem for the system were **Availability** and **Partition Tolerance (AP)**

- Ensures that the system remains accessible and responsive to users.
- Allows the system to continue operating despite network partitions.



Handling Intra-DB Consistency



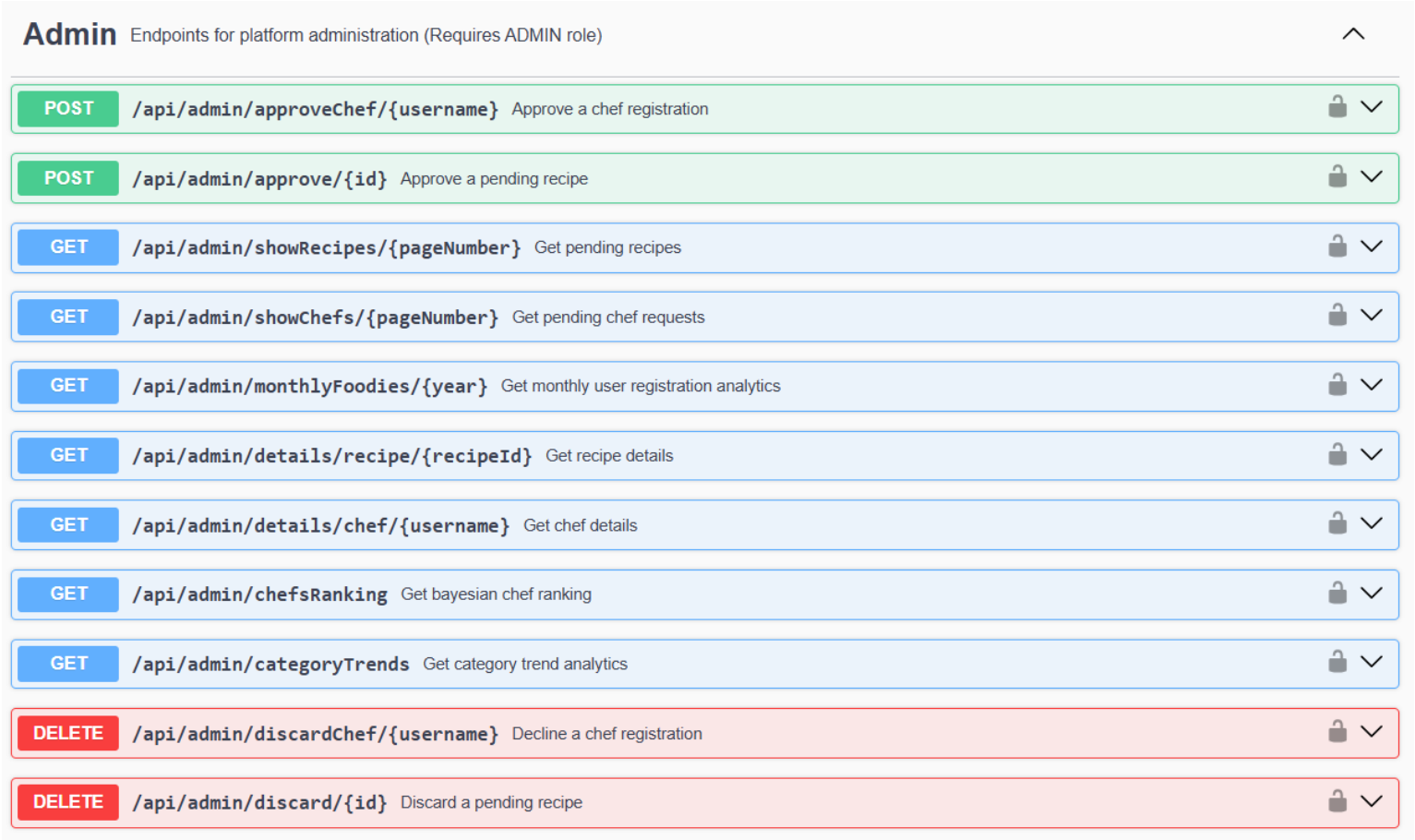
Redis cache rules

- **Add ingredient**
  Invalidate suggestion cache
- **Lazy eviction (read-repair)**
  Stale recipe persists in Redis after deletion from MongoDB/Neo4j → on access, "not found" triggers on-the-fly removal
- **TTL expiration for orphaned data**
  Deleted Foodie profile → no synchronous delete to Redis; 15-day TTL, refreshed on interaction
- **Remove ingredient**
  Update Redis Smart Fridge directly (no Neo4j traversal needed; ingredient simply removed from cached set)

Swagger UI REST APIs documentation

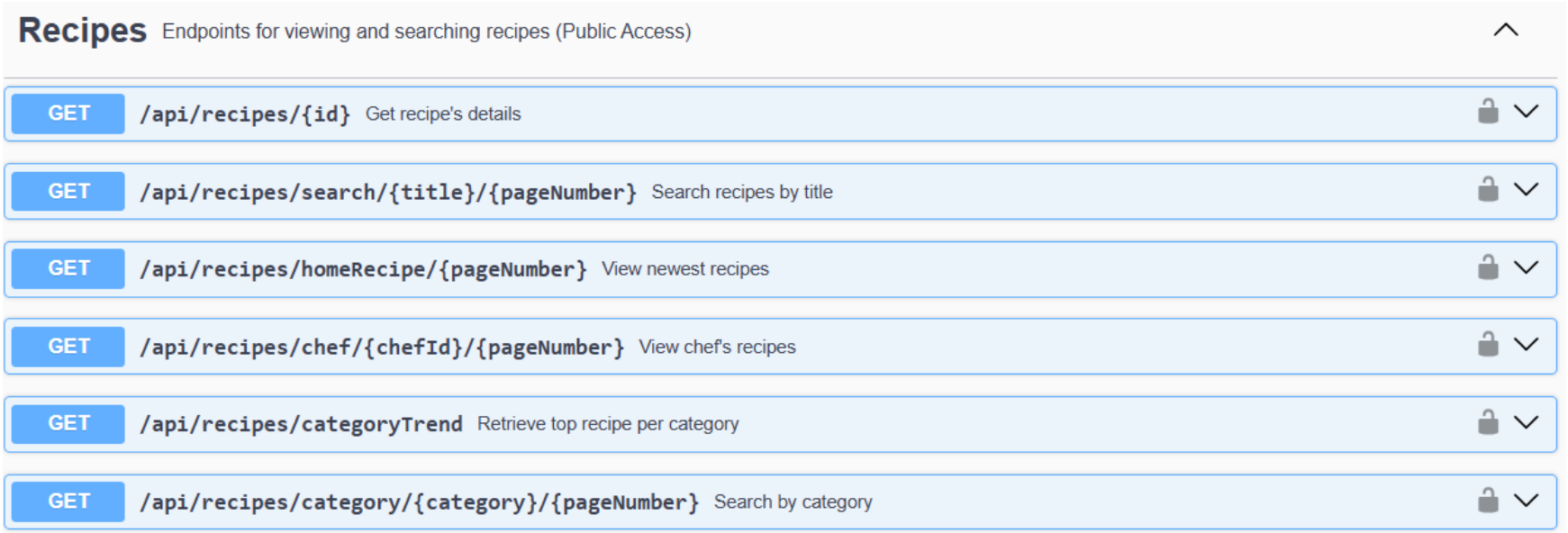
Overview

Admin



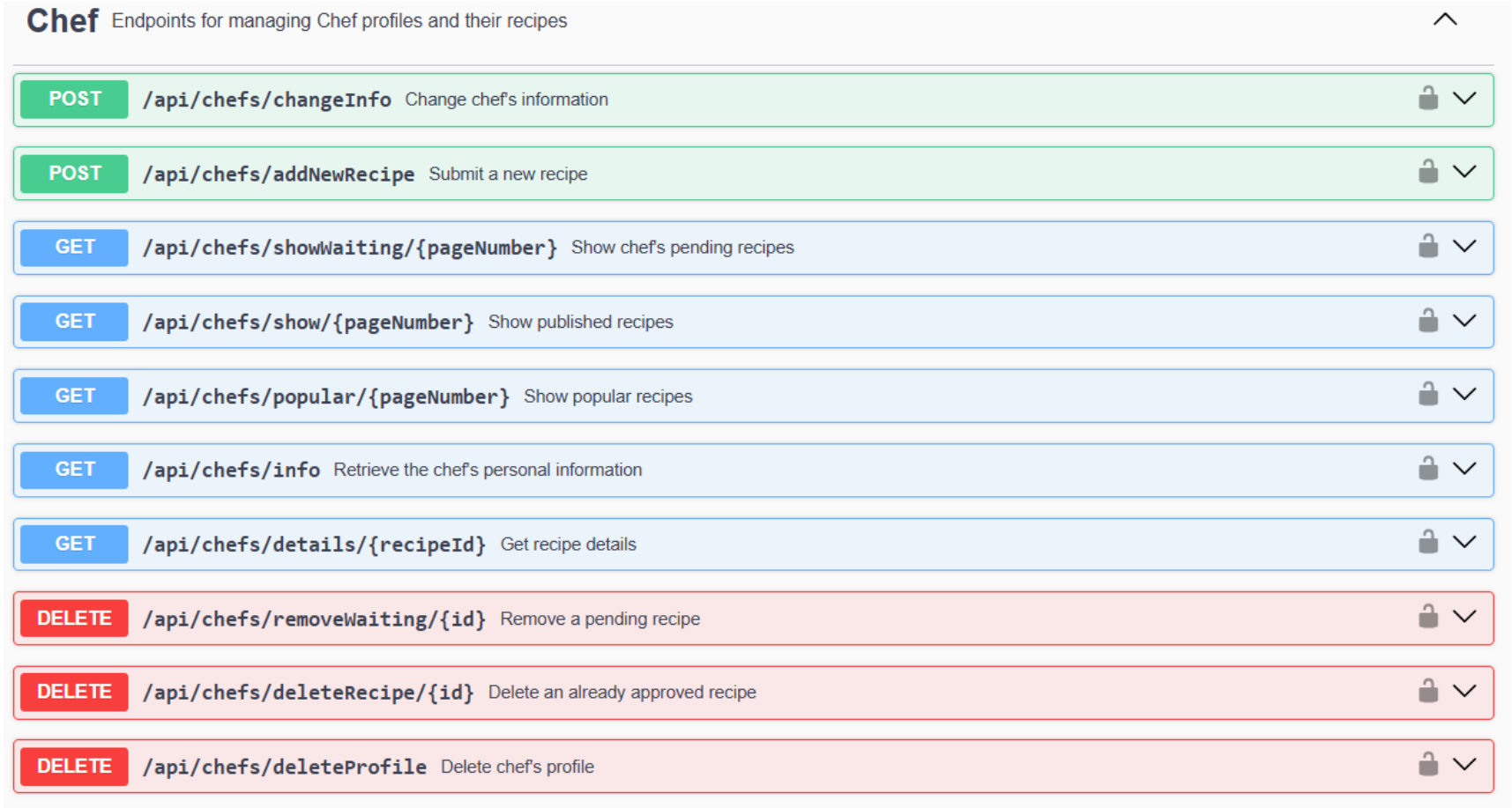
Swagger UI REST APIs documentation

Recipes



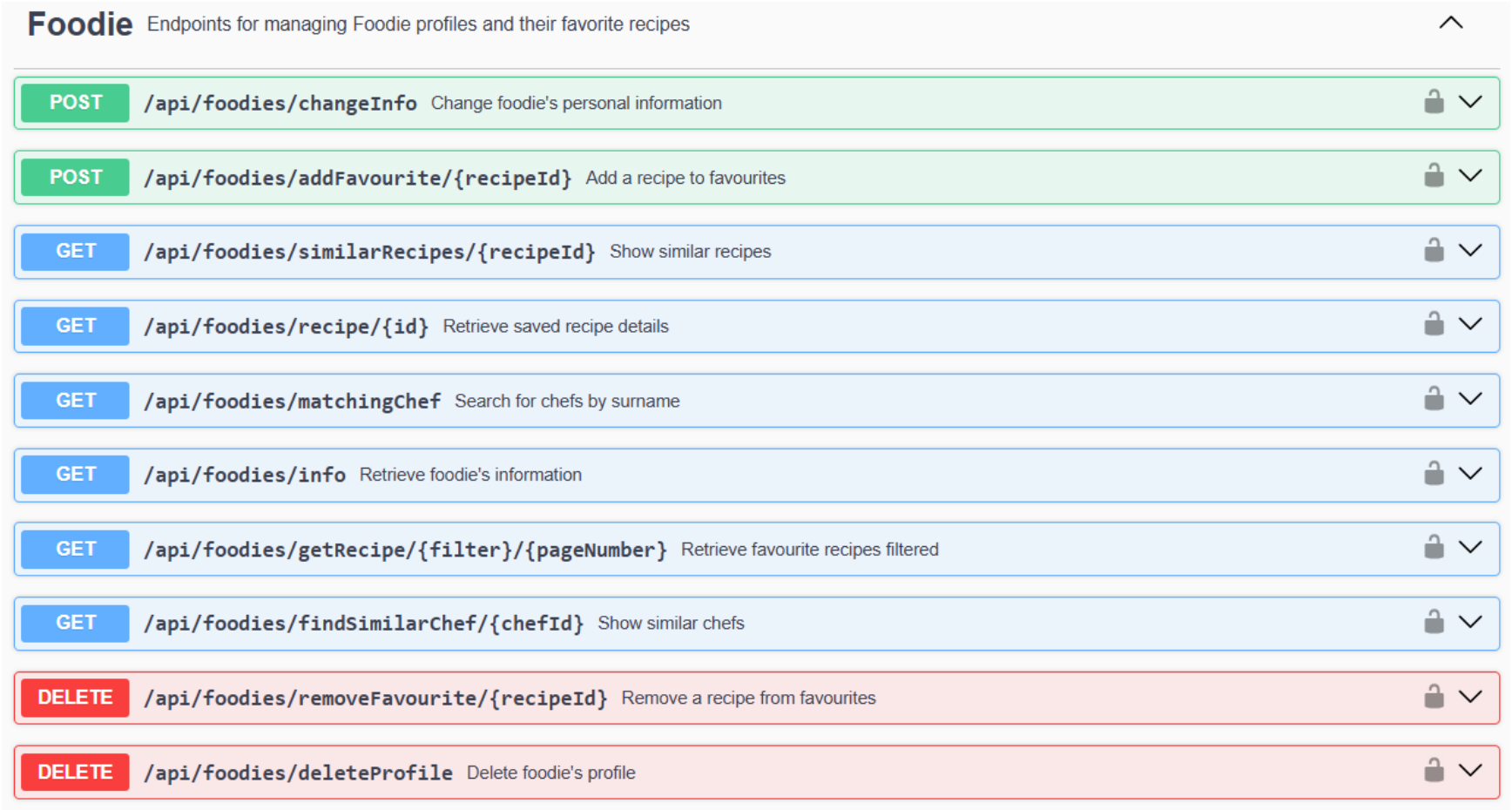
Swagger UI REST APIs documentation

Chef



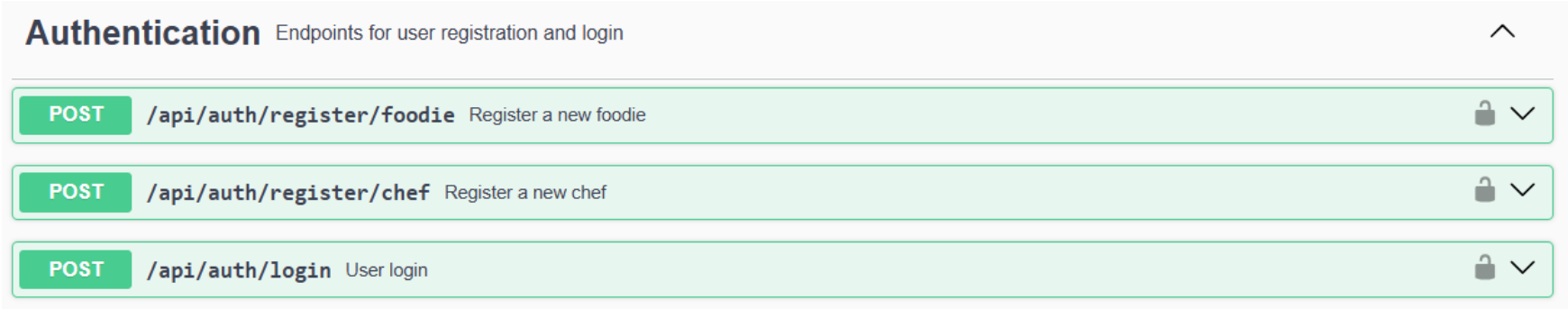
Swagger UI REST APIs documentation

Foodie

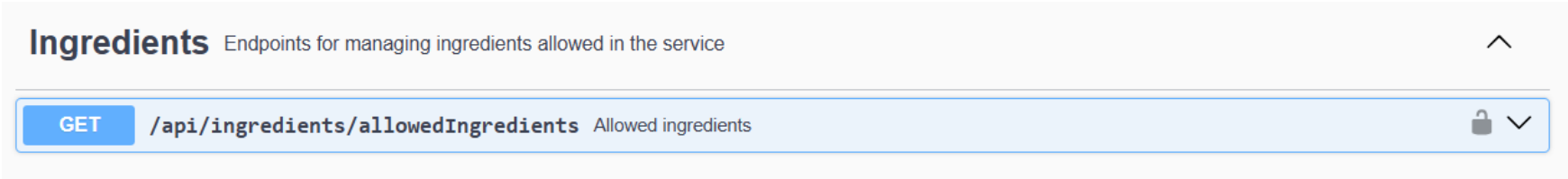


Swagger UI REST APIs documentation

Auth

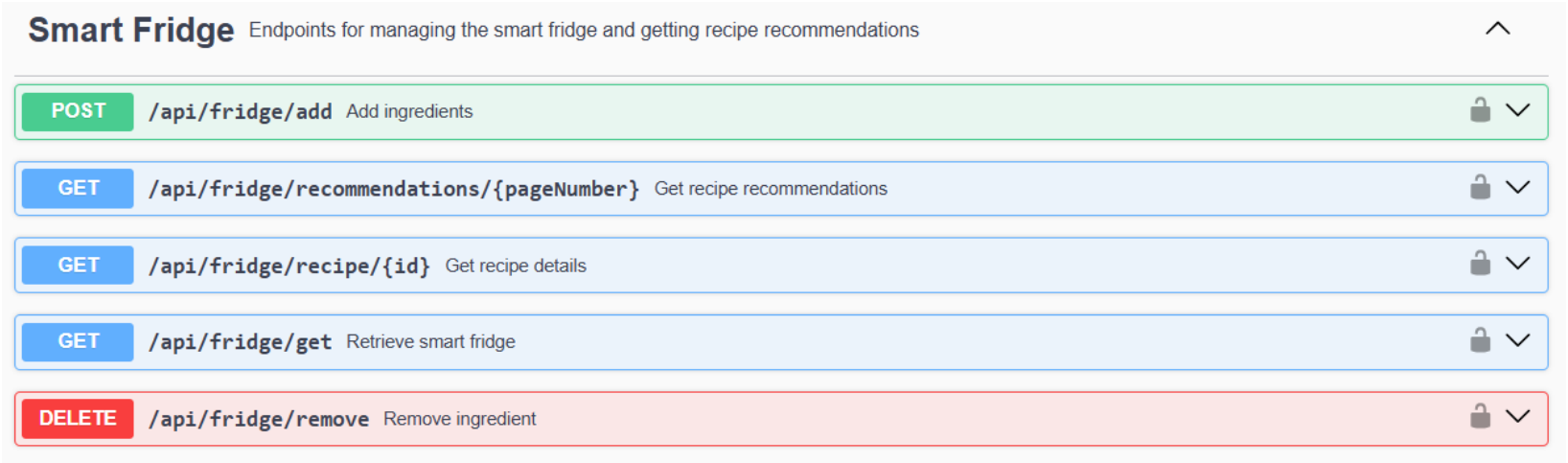


Ingredients

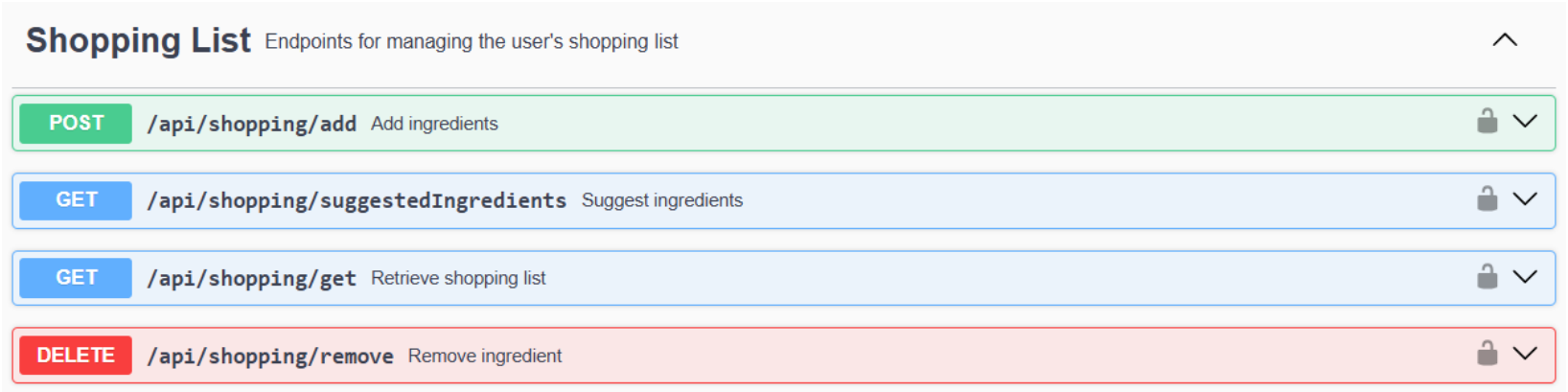


Swagger UI REST APIs documentation

SmartFridge



ShoppingList



Java doc documentation

Packages

Package	Description
it.unipi.MySmartRecipeBook	
it.unipi.MySmartRecipeBook.config	
it.unipi.MySmartRecipeBook.controller	
it.unipi.MySmartRecipeBook.dto	
it.unipi.MySmartRecipeBook.dto.recipe	
it.unipi.MySmartRecipeBook.dto.users	
it.unipi.MySmartRecipeBook.event	
it.unipi.MySmartRecipeBook.model.Mongo.ingredients	
it.unipi.MySmartRecipeBook.model.Mongo.recipes	
it.unipi.MySmartRecipeBook.model.Mongo.users	
it.unipi.MySmartRecipeBook.model.Neo4j	
it.unipi.MySmartRecipeBook.repository.Mongo	
it.unipi.MySmartRecipeBook.repository.Neo4j	
it.unipi.MySmartRecipeBook.security	
it.unipi.MySmartRecipeBook.security.jwt	
it.unipi.MySmartRecipeBook.service	
it.unipi.MySmartRecipeBook.utils.conversionFunctions	
it.unipi.MySmartRecipeBook.utils.parameters	
it.unipi.MySmartRecipeBook.utils.populateDB	

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